Mathematical Methods for International Commerce

Week 9/1: Functions of Several Variables (5.1)

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Agenda

1. Functions of Several Variables (5.1)

2. Class Activity

1. Functions of Several Variables (5.1)

Learning Objectives

- Use function notation $z = f(x, y)$
- Calculate **first-order partial derivatives**
- Calculate **second-order partial derivatives**
- Understand that $f_{xy} = f_{yx}$ under standard smoothness conditions
- Use the **small increments formula**
- Perform **implicit differentiation**

What Are Multivariable Functions?

A function with **two or more independent variables**.

Example:

Let $z = f(x, y) = 4x^2 + 3xy + y^2$

- z depends on both x and y
- This could model something like **profit**, **cost**, or **utility** depending on two inputs (e.g., labor and capital) or two goods

Multivariable Functions Graphically

A function of two variables can be visualized as a **surface** in 3D space.

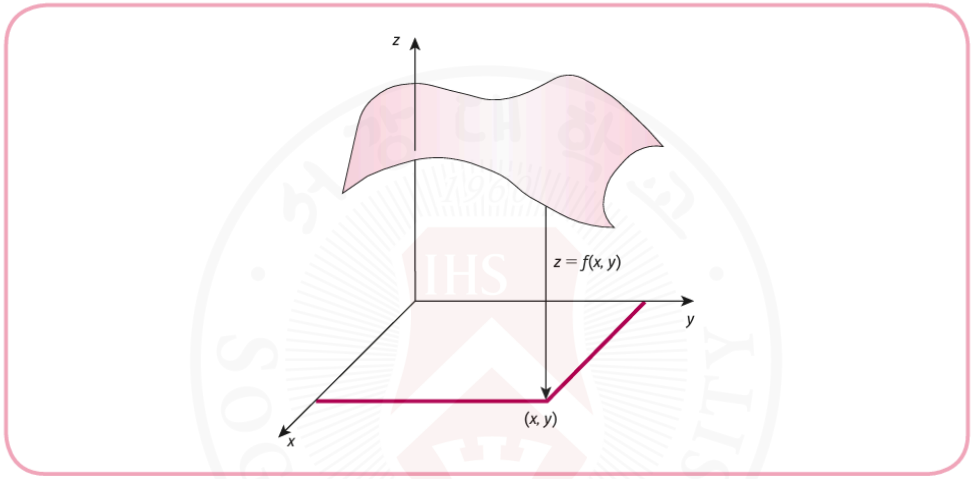


Figure 5.3

This graph shows how a function $z = f(x, y)$ maps each pair (x, y) to a height z, forming a surface.

It represents the output (e.g., cost, utility, profit) depending on two input variables.

The surface curves depending on how each variable changes, giving rise to partial derivatives.

First-Order Partial Derivatives

Partial derivative " means:

- **Partial**: only one variable is changing, the other is held constant
- **Derivative**: the rate of change of the function with respect to that variable
- **Take the derivative**: treat the other variable as a constant

Take the derivative **with respect to one variable**, holding the other constant.

Example:

Let $f(x, y) = 4x^2 + 3xy + y^2$

- $f_x = \frac{\partial f}{\partial x} = 8x + 3y$
- $f_y = \frac{\partial f}{\partial y} = 3x + 2y$

> **Interpretation**: how z changes when we slightly change x (or y) keeping the other constant.

Second-Order Partial Derivatives

Take the partial derivative **again**.

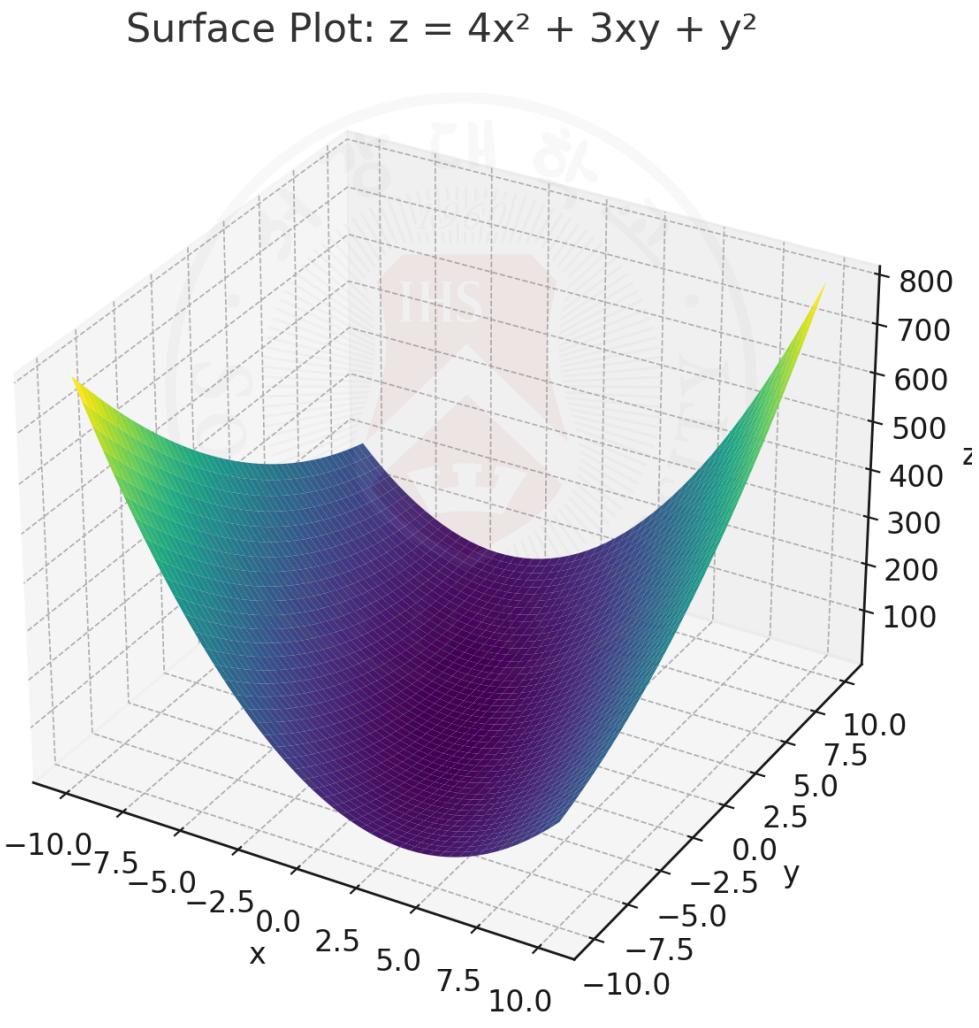
Using $f(x, y) = 4x^2 + 3xy + y^2$:

- $f_{xx} = \frac{\partial^2 f}{\partial x^2} = 8$
- $f_{yy} = \frac{\partial^2 f}{\partial y^2} = 2$
- $f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = 3$
- $f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = 3$

> $f_{xy} = f_{yx} \rightarrow$ **mixed partial derivatives are equal** (if the function has continuous second derivatives)

Surface Plot of a Multivariable Function

Function: (z = 4x^2 + 3xy + y^2)



Surface Plot of a Multivariable Function (continued)

Interpretation:

- This 3D plot shows how the output (z) varies with two input variables (x) and (y).
- The **curved surface** represents a **multivariable quadratic function**.
- As both (x) and (y) increase in magnitude, the value of (z) increases rapidly.
- The function is **convex**, indicating that it has a **minimum point** where the surface is lowest.
- Such plots are commonly used in **cost**, **utility**, and **production functions** in economics to visualize how two inputs interact.

> Convexity here means marginal effects increase as inputs grow

Small Increments Formula

Increment is a small change in the variable. Let $f(x, y)$ be a function of two variables. If we have small changes Δx and Δy, we can estimate the change in z as:

$$\Delta z \approx f_x(x_0, y_0) \cdot \Delta x + f_y(x_0, y_0) \cdot \Delta y$$

Where:

- $f_x(x_0, y_0)$ is the partial derivative of f with respect to x at (x_0, y_0)
- $f_y(x_0, y_0)$ is the partial derivative of f with respect to y at (x_0, y_0)

Example:

Let $f(x, y) = x^2 + 3y^2$, find Δz when:

- $x = 1, y = 2$
- $\Delta x = 0.1, \Delta y = -0.2$

Partial derivatives:

- $f_x = 2x \Rightarrow f_x(1, 2) = 2$
- $f_y = 6y \Rightarrow f_y(1, 2) = 12$

$\Delta z \approx 2(0.1) + 12(-0.2) = 0.2 - 2.4 = \boxed{-2.2}$ This is a linear approximation of the function.

Implicit Differentiation

When variables are related **implicitly** (not as $y = f(x)$), we still find $\frac{dy}{dx}$.

Example:

Let $x^2 + y^2 = 25$. Differentiate both sides:

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0 \Rightarrow 2x + 2y \cdot \frac{dy}{dx} = 0$$

Solve:

$$\frac{dy}{dx} = -\frac{x}{y} \quad \text{We treat } y \text{ as a function of } x$$

Practice Problems

1. Let $f(x, y) = 5x + xy^2 - 10$, and $g(x_1, x_2, x_3) = x_1 + x_2 + x_3$. Evaluate:

   - (a) $f(0, 0)$
   - (b) $f(1, 2)$
   - (c) $f(2, 1)$
   - (d) $g(5, 6, 10)$
   - (e) $g(0, 0, 0)$
   - (f) $g(10, 5, 6)$

2. Find f_x, f_y, f_{xx}, f_{yy}, f_{xy} for $f(x, y) = x^2y + y^3$

3. Use small increments: $f(x, y) = 2x + y^2$, $x = 2$, $y = 1$, $\Delta x = 0.05$, $\Delta y = -0.1$

4. Implicit diff: Given $x^2 - xy + y^2 = 7$, find $\frac{dy}{dx}$

Summary

- Partial derivatives help analyze functions with multiple inputs
- Second-order derivatives are used to analyze curvature and optimization
- Small increments formula estimates change efficiently
- Implicit differentiation handles **non-solved** functions

In economics, these techniques are used in:

- **Cost functions** with multiple inputs
- **Utility and production functions**
- **Marginal analysis** in multivariate cases

2. Group Activity: Cost Function Strategy Game

Objective:

Use teamwork to analyze how changes in labor and capital affect total cost and marginal cost.

Instructions:

- Form **6 groups** of **4-5 students**.
- Each group gets:
  - A cost function: $C(L, K) = 20L + 30K + LK$, where The cross term LK creates increasing marginal costs
  - A table with sample values for L and K.
- Tasks:
  1. Calculate C_L and C_K (partial derivatives).
  2. Interpret their economic meaning.
  3. Discuss in your group: **How would increasing L while holding K constant affect costs?**
  4. Sketch a contour (level curve) or grid interpretation.

Each group will present a 2-minute explanation of:

- Your calculated derivatives
- Your insights on labor and capital usage

Sample Table:

L (Labour)	K (Capital)	C(L, K)
1	1	51
2	1	72
1	2	82
2	2	104
3	3	177

Next Classes

- (April 30) Partial Elasticity and Marginal Functions (5.2)

Any QUESTIONS?

Thank you for your attention and participation!